

Lab Report-2

1. Write a C/C++ program that implements Magic Square, Stack, and Queue.

Objectives:

- To create a Magic Square using a C/C++ program.
- To implement a Stack using an array.
- To perform basic Stack operations like push, pop, and display.
- To implement a Queue and perform operations like enqueue, dequeue, and display.

Magic Square:

Source code:

```
KAISER
Update
MagicSquare.cpp
Report > LabReport2 > MagicSquare.cpp > main()
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int n,i,j,k,l,key;
5     cout<<"Enter an odd number: "<<endl;
6     cin>>n;
7     int square [n][n];
8     if(n%2==0){
9         cout<<"n is Even";
10        return 0;
11    }
12    for(i=0;i<n;i++){
13        for(j=0;j<n;j++){
14            square[i][j]=0;
15        }
16    }
17    int mid=(n-1)/2;
18    square[0][mid]=1;
19    j=mid;
20    for(key=2;key<=n*n;key++){
21        if(i>0){
22            k=i-1;
23        }
24        else{
25            k=n-1;
26        }
27        if(j>0){
28            l=j-1;
29        }
30        else{
31            l=n-1;
32        }
33        if(square[k][l]>0){
34            i=(i+1)%n;
35        }
36        else{
37            i=k;
38            j=l;
39        }
40        square[i][j]=key;
41    }
42    for(i=0;i<n;i++){
43        for(j=0;j<n;j++){
44            cout<<square[i][j]<<" ";
45        }
46        cout<<endl;
47    }
48 }
```

Input & Output:

```
cd "/Users/macbookairm5/Documents/KAISER"
p -o MagicSquare && "/Users/macbookairm5/Documents/KAISER"
macbookairm5@KAISER KAISER % cd "/Users/macbookairm5/Documents/KAISER"
port2/" && g++ MagicSquare.cpp -o MagicSquare
Lab Report/LabReport2/"MagicSquare
Enter an odd number:
5
15 8 1 24 17
16 14 7 5 23
22 20 13 6 4
3 21 19 12 10
9 2 25 18 11
macbookairm5@KAISER LabReport2 %
```

Stack:

Source code:

```
Algorithm Lab Report > LabReport2 > Stack.cpp > main()
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int i,n,a[20],max=4,top=-1;
5      while(true){
6          cout<<"Enter 1 for insertion\n";
7          cout<<"Enter 2 for deletion\n";
8          cout<<"Enter 3 for display\n";
9          cin >> n;
10         if(n==1){
11             if(top==max)
12                 cout<<"Stack is overflow\n";
13             else{
14                 int b;
15                 cout<<"Enter the value\n";
16                 cin>>b;
17                 top=top+1;
18                 a[top]=b;
19                 cout<<"Data successfully inserted\n";
20             }
21         }
22         else if(n==2){
23             if(top==--1)
24                 cout<<"Stack is underflow\n";
25             else{
26                 cout<<"Deleted value is "<<a[top]<<"\n";
27                 top = top - 1;}
28         }
29         else if(n==3){
30             if(top==--1)
31                 cout<<"Stack is empty\n";
32             else{
33                 cout<<"The present values are:\n";
34                 for(i=0;i<=top;i++)
35                     cout<<a[i]<<endl;
36                 cout<<endl;}
37         }
38         else{
39             break;
40         }
41     }
42     return 0;
43 }
```

Input & Output:

```
cd "/Users/macbookairm5/Documents/KAISER/Algorithm Lab Report/
Lab Report/LabReport2/"Stack
macbookairm5@KAISER KAISER % cd "/Users/macbookairm5/Document
irm5/Documents/KAISER/Algorithm Lab Report/LabReport2/"Stack
Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
1
Enter the value
5
Data successfully inserted
Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
3
The present values are:
5

Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
2
Deleted value is 5
Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
3
Stack is empty
Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
```

Queue:

Source code:

```
Algorithm Lab Report > LabReport2 > Queue.cpp > ...
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int i,n,a[20],front=-1,rear=-1,max=4;
5      while(true){
6          cout<<"Enter 1 for insertion\n";
7          cout<<"Enter 2 for deletion\n";
8          cout<<"Enter 3 for display\n";
9          cin >> n;
10         if(n==1){
11             if(rear==max)
12                 cout<<"Queue is overflow\n";
13             else{
14                 int b;
15                 cout<<"Enter the value\n";
16                 cin>>b;
17                 if(front==max)
18                     front=0;
19                 rear=rear+1;
20                 a[rear]=b;
21                 cout<<"Data successfully inserted\n\n";
22             }
23         }
24         else if(n==2){
25             if(front==max || front>rear)
26                 cout<<"Queue is underflow\n\n";
27             else{
28                 cout<<"Deleted value is "<<a[front]<<"\n";
29                 front=front+1;
30             }
31         }
32         else if(n == 3){
33             if(front == max || front>rear)
34                 cout<<"Queue is empty\n\n";
35             else{
36                 cout<<"The present values are:\n";
37                 for(i=front;i<=rear;i++)
38                     cout<<a[i]<<endl;
39                 cout<<endl;
40             }
41         }
42         else{
43             break;
44         }
45     }
46     return 0;
47 }
```

Input&Output:

```
cd "/Users/macbookairm5/Documents/KAISER/Algorit
/"Queue
macbookairm5@KAISER KAISER % cd "/Users/macbooka
gorithm Lab Report/LabReport2/"Queue
Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
1
Enter the value
56
Data successfully inserted

Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
1
Enter the value
45
Data successfully inserted

Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
3
The present values are:
56
45

Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
2
Deleted value is 56
Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
3
The present values are:
45

Enter 1 for insertion
Enter 2 for deletion
Enter 3 for display
█
```

Conclusion:

In this experiment, a Magic Square was successfully generated using a C/C++ program. The logical arrangement and pattern of numbers were clearly understood through this implementation. A Stack was also implemented using an array, and its basic operations such as push, pop, and display were performed successfully. Similarly, a Queue was implemented, and operations like enqueue, dequeue, and display were carried out. Through these programs, the concepts of basic data structures and their operations were clearly understood, and programming skills were improved.